When the machine works like human is called / learn the machine / AI is behave like human fn.

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31-7-24 Artifical Intelligence The technique used is M.L

→ Artifical Intelligence :- AI is refers to intelligent machine that can mimic human cognitive function like learning, decision making and problem soluing.

→ Goal of AI (Artifical Intelligence)

(i) Reasoning
(ii) Learning
imp (iii) Decision Making
(iv) Problem solving

raw data → prepare the data
↓
then apply Mach Pro learning

→ How does AI Work?

AI enables machine to learn from data and recognized pattern to perform task more efficiently and effectively. AI work in the following steps :-

(1.) Input data (Data is the most imp part of the process) Data is use to work AI.
Data is collected from various sources and its sorted into categories.

(2.) Processing
clustering.
AI sort the data using pattern it has been programed to learn until it recognise similar pattern in the data.

(3) Outcomes
AI can then use those pattern to predict outcomes.

→ AI machines learn from their own mistakes. (this is the reason AI become powerful.

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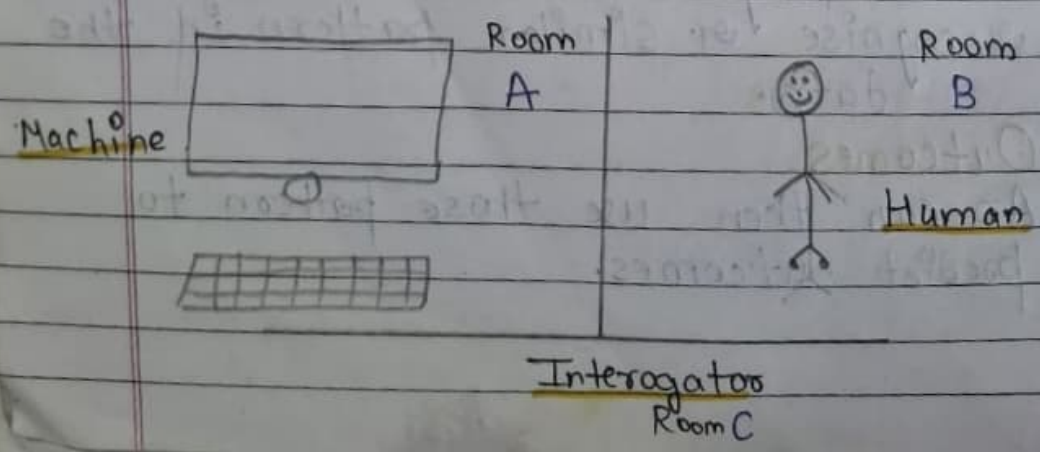
AI is possible due to Data.

(4.) Adjustment
If a data sets considered a fail AI learn from that's mistake and the process is repeated again under different conditions.

(5) Assessment
In this way AI is consistently learning and improving.

Turing Test
→ This Test was developed by Alan Turing a computer scientist in 1950 He proposed that the turing test is used to determine wheater or not a computer or machine can think intelligently like human.

→ The Turing Test is a widely used major of a machine ability to demonstrate human like intelligence.



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History of AI

→ 1950 :- In 1950, a computer scientist Alan Turing publish a paper computing machinery and intelligence which introduce the turing test a major of computer intelligence.

→ 1956 :- A significant turning point of AI research occur in 1956 when John Mc Carthy (father of AI) first uses the frace of artifical intelligence at the Dart mouth Workshop.

→ 1959 :- In 1959, first AI Lab was establis in MIT.

→ 1960 :- In 1960, first robot was introduc which was general motor robo. (prototy p

→ 1961 :- In 1961, first chatbots goes invented which is known as Eliza. (Josu Scien

→ 1972 :- In 1972, the first intelligent humanoid robot was built in Japan. which wa name as WABOT- I.

→ 1980 :- In 1980, expert system (decide some rules) was programmed that ammulate the decisio making ability.

(International Business Machine)

→ 1997 :- In 1997, IBM's deep blue beats the world champion Garry kasparov in the game of chess

→ 2005 :- In 2005, standford racing team autotomaous robotic cars STANLEY wins the DARPA Grand Racing challenge.

→ 2006 :- AI comes in business world till the year 2006. Companies like facebook Twitter, Netflix also started using AI

→ 2011 :- IBM's question, answer system WATSON develop in 2011.

→ 2016 :- Hongkong developed the humanoid robot Sophia in 2016.

→ 2022 :- In 2022, chatgpt a large language model based chatbots introduce on november 30.

image videos ← Visual data and processing in visual data

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Artifical Intelligence Techniques:-

(i) Machine learning (learn from data pattern)
A sub field of AI uses algorithum to enable systems to learn from data and make predictions or decision making without being explicity programs.

(ii) NLP (Natural language Processing)
A branch of AI that focus on enabling computers to understand, interpret and generate human language.

(iii) Computer Vision
A field of AI that deals with the processing and analyses of visual information using computer algorithum.

(iv) Robotics
AI powered robots and automation system that can perform task in manufacturing, healthcare, rital and in the other industry.

(v) Expert System
AI system that mimic the decision making ability of a human expect in a specific field.

→ fuzzy logic technique of ai - that is used to take decision.

example:

(vi) Chatbots

AI power virtual assistance that can interact with user through text based or voice based interface.

calling customer care - virtual assitance

all apps have Chatbots they provid every query.

Advantages and Limitation of AI

→ Managment, productivity and effectiveness of things have increased. presiged

→ It will available 24x7.

→ Personalisation and Costomisation will increased.

→ Security in

→ Using AI mobile have face recoginition lock

→ Using AI house have face recoginition lo

→ Using AI in companies it have face recognition of attendance.

→ Human life becomes easy by using AI.

Manufacturing Sector

→ Increase productivity

→ Using less power

→ Many power using less

→ automation

Healthcare

→ Making robots surgery

→ Using machine learning we can analyses complex diseases. (by using IMR image. image processing

→ Cancer at early stage can be detected.

Transport

→ using metro.
→ proper information signal.
→ proper information of traffic.
→ self drive and automation cars

Finance

→ predict market
algorithum bared prediction
→ prediction of fraud of companies
→ using AI, they analyses data.
→ prediction of stock market.

Education

→ availabletiy AI assistance.
→ Decision making becomes easy.
→ Suggestions

Jobs

→ Cration of Jobs (data-analyst, data scientist)

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Limitations :-

(i) High cost (expensive) - learning machine will take high cost.

(ii) Human dependency and lazyness

(iii) Privacy Concern :- concern about the private data.

(iv) Job displacement / replacement
There many jobs displacement (customer care, data-entry operator)

(v) Can't think beyond the limit
:- it only think and do work on the limits.

- Job displacement :- Human wages jobs like customer care, repetitive jobs are gone

- AI's cannot think beyond the limit, they cannot go after the limits, are not much suitable for certain sectors.

Imp ★ Applications of AI :-

- Home service robot : like mopping the floor.
- Alexa and siri helps a lot answering queries
- Transport : self driving cars.
- Healthcare : Robotic surgeries are ~~p ors~~ possibles, can detect complex disease by images.
- Education : A z tutors, assistance are available for guidance and information
- Public security and safety : Facial recognition
- Employment and workspace : Data Science jobs, data analyse job, data collection job, data connection job are available.
- Entertainment : Netflix, personality choices accordingly, recommendation are available.
- Finance and Banking Insurance :
  Market prediction
  Trading based algorithm
  Fraud can be detected for like bank loans.

- Impact of AI :-

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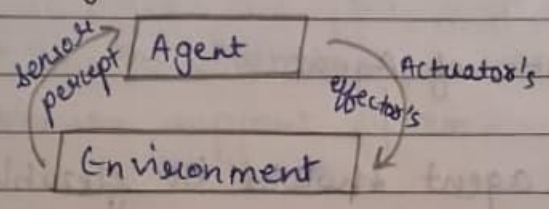
→ Both advantage & limitations will be written here

* Example of AI :-

→ Alexa, Siri
→ Automation Cars
→ Facial recognition
→ Home service robots.

* Intelligent Agent :-



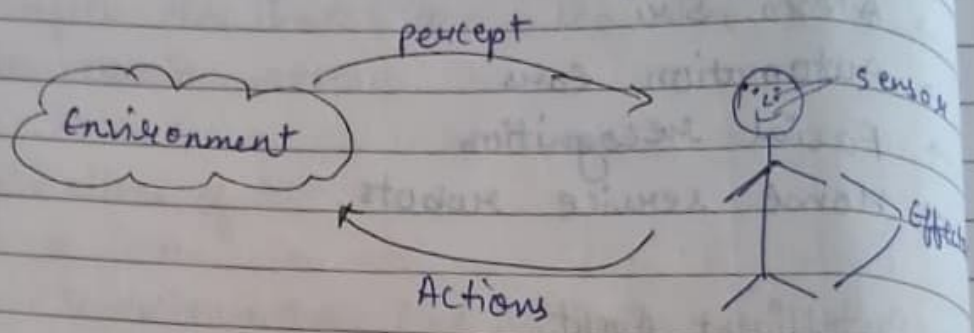
Intelligent Agent Structure

An agent is anything like human, computer program, robot or any system that is designed to percieve its environment, make decision & take action to achieve a specific goal or set of goals.
The agent operate autonomously, meaning it is not directly controlled by a human

* Nature of Agent

• Reactive Agent :- Intelligent agent respond to immediate senses from the environment and take action based on those senses.
example :- tic tac toe

- Proactive Agent :- Those Intelligent agent who take initiative and plan ahead to achieve their goal.
  example :- Autonomous car or selfdriving car



★ Characteristic of Agent

- Intelligent agent should be flexible to change its environment
- Intelligent agent taking goal directed initiative when appropriate
- Learning from its own experience, its environment and interaction with others
- The agent is capable of interacting in a peer to peer manner with other agents or humans

★ Goals of Agents :-

- High performance :
- Optimised result :
- Rational action :

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→ Flowchart

Agent $\xrightarrow{\text{Percept}}$ Environment → Action → change in environment

* Rational Agent :- Rational Agent in AI is an agent that perform action to achieve the best possible outcomes based on its perception and knowledge. It operate under the premises of rationality where it consistently makes decision that maximize its performance ~~major~~ measure

example :- In the robotics, automated trading system, decision support system.

* Performance Measure in Rational Agent :- Performance measure of rational agent is the criteria which determine how success agent is.

→ Agent has to take initiative to perciere info. from the environment
→ Agent has to generate a sequence of action according to the percept it recieve.
→ The sequence of action causes the environment to go to a sequence of state.
→ If the sequence is desirable then the agent has performed well.

PEAS
P → Perception
E → Environment
A → Action
S → Sensor

} dependance upon

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* Task Environment & its properties

An environment in AI is the surrounding of the agent, the agent take input from the environment through sensors & ~~delivers~~ deliver the output to the environment through actuators and factors.

① Fully observable and partially observable
example chess driving

② Deterministic and stochastic (Random)
Predict ~~kare~~ things happen
skte ho environment randomly

③ Competitive and Collaborative
compitition collab to achieve
to achieve goal enviro goal environment

④ Single Agent & Multi agent
single many

⑤ Dynamic & static
that changes that doesn't
frequently change

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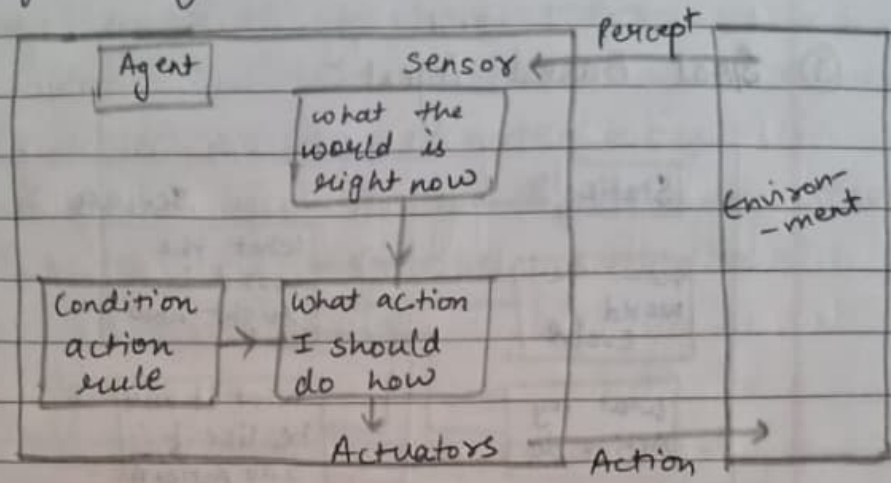
* Types of Agents :-

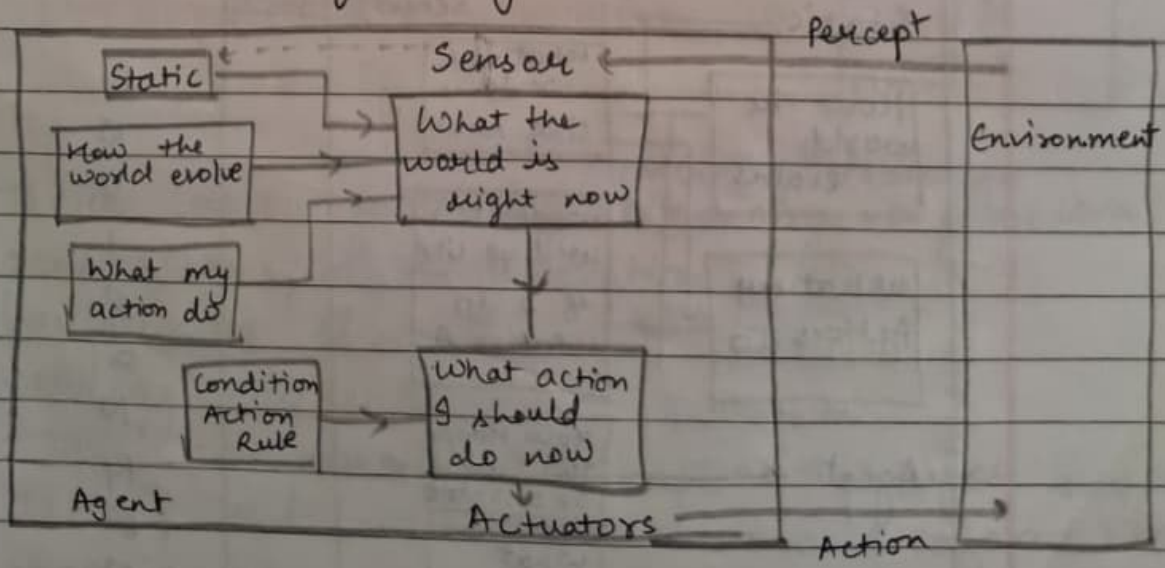
1 Simple Reflex agent :-



Key point

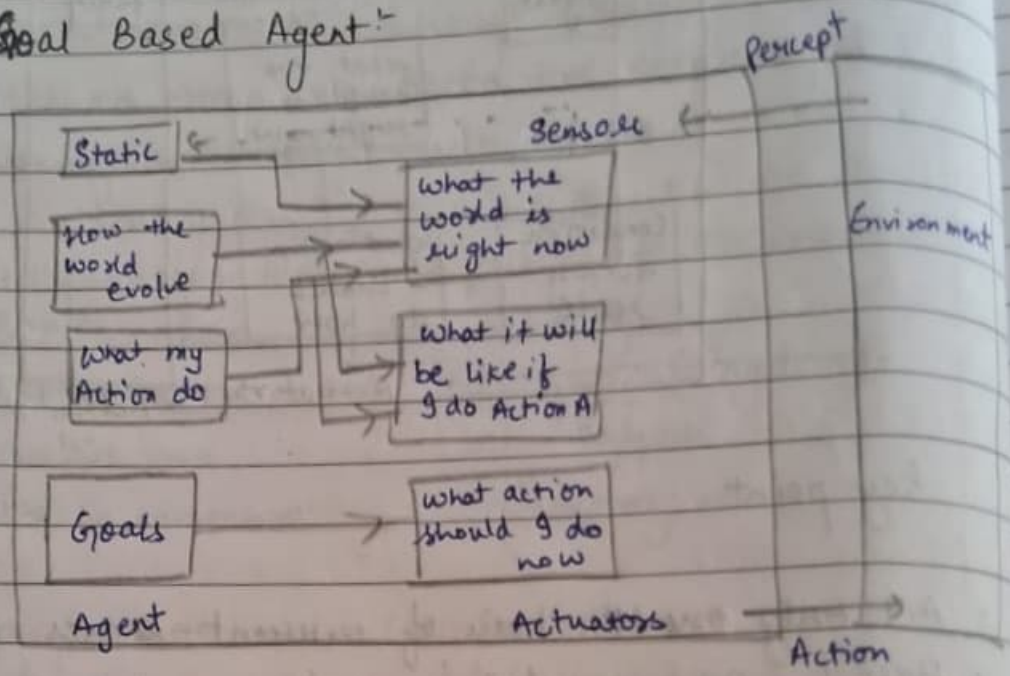
- Act only on the basis of current senses
- Based on condition rule
- Ignore the rest of the historic perception.
- Environment should be fully observed.

2 Model Based Reflex Agent :-



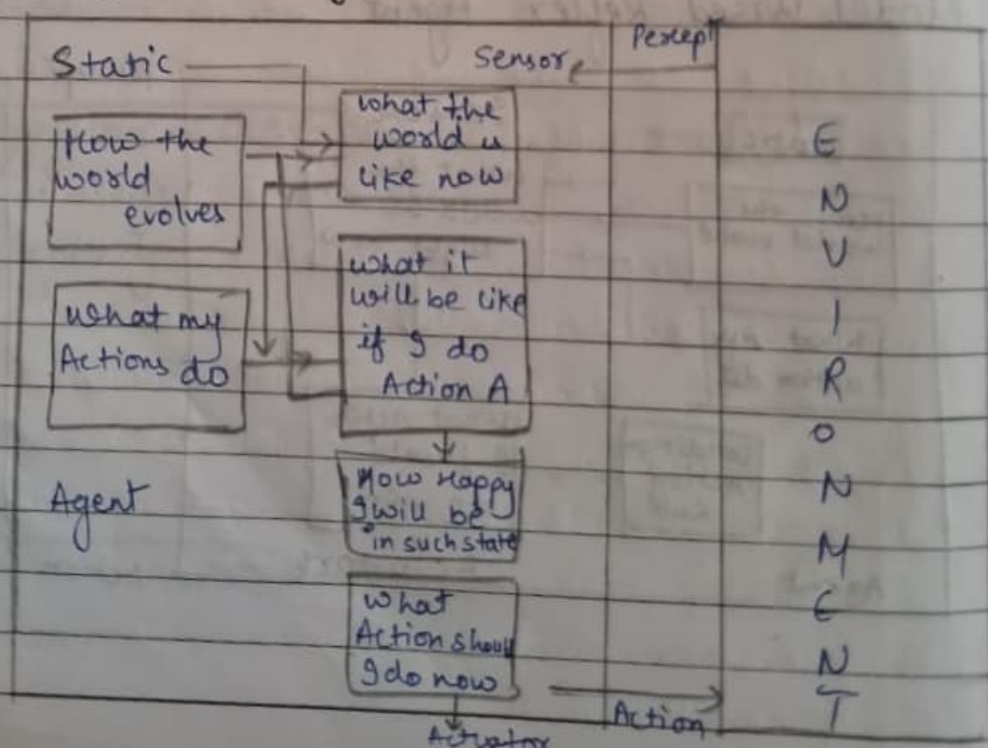
- Environment is partially observed (works in)
- It stores the perception history / it has its own internal model

③ Goal Based Agent:-



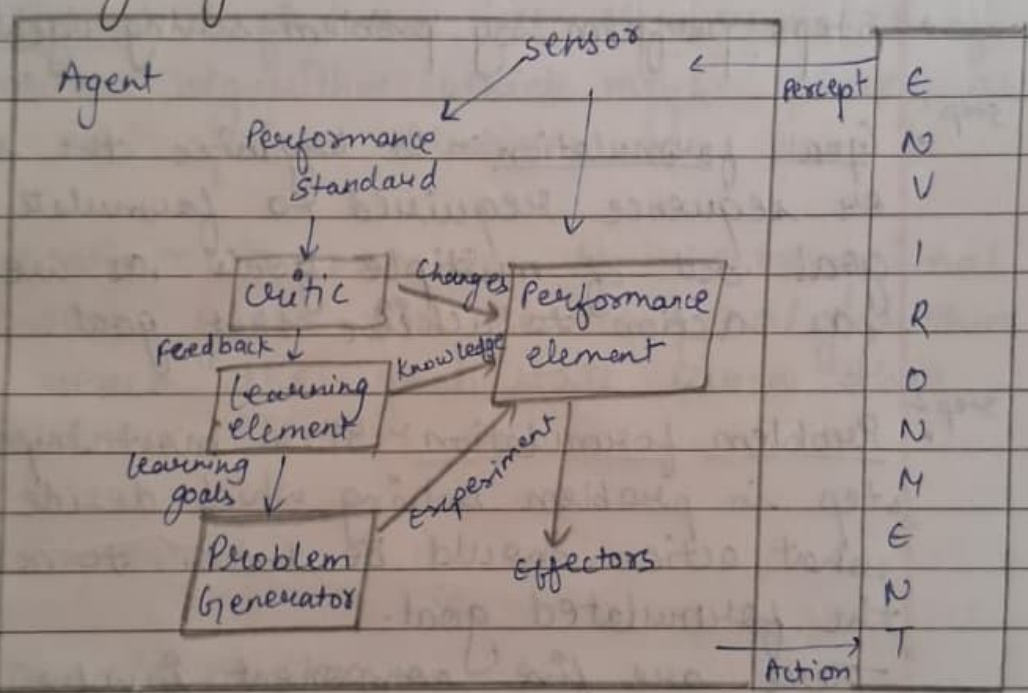
- This agent only take goal oriented action

★ Utility Based Agent:-



- This agent focus on the utility not on the goal.
- This agent uses utility function which is a mathematical representation which evaluate the desire output of various event or state. It is denoted by U.
- This agent deals with happy & unhappy state
- This agent work in partial environment.

(5) Learning Agent



- There are following ~~elem~~ element of learning agent ~~problem~~
- ~~generation~~ problem generation
- performance element
- learning element

→ This component is responsible for suggesting actions that will lead to new & informative experience